

KALKAR, Germany

WMO: 104040

Lat: 51.750N

Lon: 6.283E

Elev: 141

StdP: 14.62

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	22.6	12.9	10.6	20.7	17.1	13.1	24.4	26.0	47.8	22.6	47.7	4.9	70	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	86.6	67.9	82.6	66.3	78.8	64.7	69.8	82.4	68.0	79.8	66.2	76.4	5.4	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	94.7	74.9	63.6	88.6	72.7	62.0	83.6	71.2	33.8	82.3	32.3	79.6	30.9	76.4	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	15.8	13.5	13.5	94.0	6.0	2.6	9.2	95.8	5.7	97.4	2.3	98.8	-2.0	100.7
			13.0	74.2	5.9	1.5	8.8	75.2	5.3	76.1	2.1	76.9	-2.2	78.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	37.0	39.2	43.9	50.6	57.3	62.1	65.7	65.7	59.8	52.3	44.4	37.6	(6)
		DBStd	12.23	8.02	7.44	6.40	6.73	6.46	6.08	5.88	5.52	5.20	6.07	6.32	7.64	(7)
		HDD50	1619	404	305	205	74	12	0	0	1	46	184	388		(8)
		HDD65	5210	867	722	654	434	251	127	62	57	173	394	618	851	(9)
		CDD50	2118	2	3	16	91	238	364	487	487	294	117	16	3	(10)
		CDD65	234	0	0	0	1	12	41	84	79	17	0	0	0	(11)
		CDH74	2127	0	0	0	35	154	395	767	663	111	2	0	0	(12)
		CDH80	652	0	0	0	4	29	111	262	227	19	0	0	0	(13)
(14)	Wind	WSAvg	5.8	7.4	7.4	6.5	5.6	5.1	4.7	4.9	4.6	4.8	5.9	6.3	7.0	(14)
(15)	Precipitation	PrecAvg	32.6	2.8	2.5	2.2	1.7	2.5	2.9	3.2	3.4	2.7	2.7	3.0	3.1	(15)
(16)		PrecMax	41.9	5.2	5.1	4.8	3.5	5.3	7.1	6.1	8.5	6.6	8.1	4.6	6.5	(16)
(17)		PrecMin	25.2	0.2	0.6	0.4	0.0	0.5	0.4	0.8	0.9	0.4	0.8	0.3	1.0	(17)
(18)		PrecStd	4.1	1.3	1.2	1.3	0.8	1.1	1.6	1.4	2.0	1.6	1.3	1.3	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	55.2	57.2	66.0	78.8	83.9	89.3	91.9	92.0	82.9	71.8	61.1	55.8		(19)
(20)		0.4% MCWB	50.9	49.9	53.7	58.5	65.7	69.8	69.4	70.0	66.4	62.5	55.3	52.4		(20)
(21)		2% DB	52.4	53.7	60.9	72.2	78.2	82.8	86.7	86.0	77.1	66.7	57.6	53.2		(21)
(22)		2% MCWB	49.1	48.7	51.9	57.1	61.1	66.6	67.9	68.1	65.2	59.7	53.9	50.4		(22)
(23)		5% DB	50.1	51.6	57.0	67.3	74.2	78.4	82.2	81.3	72.4	63.5	55.3	50.6		(23)
(24)		5% MCWB	46.9	47.4	49.9	54.5	59.4	64.6	66.6	66.5	62.3	57.9	52.3	47.9		(24)
(25)		10% DB	47.7	49.3	53.6	62.7	70.0	74.2	77.6	76.7	68.6	60.9	53.0	48.3		(25)
(26)		10% MCWB	44.9	45.9	48.2	52.7	57.9	62.2	65.1	64.7	60.2	56.4	50.0	45.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	52.2	51.5	55.4	61.2	67.3	71.6	72.8	72.4	68.5	63.3	56.3	53.1		(27)
(28)		0.4% MCDB	54.0	54.5	62.8	75.6	80.3	85.1	86.3	86.9	78.6	69.7	59.4	55.2		(28)
(29)		2% WB	49.5	49.6	53.0	58.1	63.9	68.2	69.7	69.6	65.7	60.8	54.4	50.6		(29)
(30)		2% MCDB	52.1	52.7	59.1	69.1	74.4	80.3	81.6	82.6	74.7	65.8	57.4	52.8		(30)
(31)		5% WB	47.1	47.9	50.7	55.7	61.1	65.7	67.9	67.5	63.5	58.8	52.5	48.3		(31)
(32)		5% MCDB	49.9	51.0	55.8	65.3	70.2	75.7	79.7	78.7	70.9	62.6	55.2	50.3		(32)
(33)		10% WB	44.7	46.0	48.5	53.5	58.9	63.4	65.8	65.5	61.3	56.9	50.3	45.9		(33)
(34)		10% MCDB	47.4	49.2	53.2	61.9	68.0	72.2	76.0	74.9	67.5	60.5	52.6	48.3		(34)
(35)	Mean Daily Temperature Range	MDBR	8.5	10.1	13.5	17.6	17.9	18.2	17.5	17.9	15.6	13.1	9.6	8.0		(35)
(36)		5% DB MCDBR	9.6	12.3	19.2	24.8	25.3	25.1	24.9	25.3	21.0	15.7	11.4	9.0		(36)
(37)		5% DB MCWBR	8.1	9.2	12.3	13.1	12.8	12.8	11.2	11.6	11.7	10.6	9.2	8.2		(37)
(38)		5% WB MCDBR	9.4	10.4	16.7	21.9	21.9	21.8	22.4	22.2	18.2	14.6	10.8	8.9		(38)
(39)		5% WB MCWBR	8.4	8.5	11.3	12.6	12.9	12.4	11.1	11.5	11.1	10.7	9.1	8.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.353	0.382	0.416	0.420	0.418	0.429	0.420	0.394	0.366	0.344	0.325		(40)
(41)		taud	2.362	2.334	2.256	2.199	2.227	2.250	2.226	2.273	2.335	2.394	2.370	2.338		(41)
(42)		Ebn at Noon	212	240	254	259	265	266	260	256	249	232	204	191		(42)
(43)		Edh at Noon	20	27	35	41	42	41	42	38	32	25	20	17		(43)
(44)	All-Sky Solar Radiation	RadAvg	228	441	801	1292	1567	1679	1597	1354	987	580	270	175		(44)
(45)		RadStd	31	83	116	142	136	139	198	121	110	52	32	22		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (84 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page